

Experiment 3 Convolution (Convolution Sum) - Part II

I. I. Experiment Objectives

1. Understand the concepts of sampling and time delay;
2. Examine how convolution represents the system response as a superposition of impulse responses.

II. Pre-Lab Preparation

1. Review the operation of pulse generators and sequence generators;
2. Review the definition of convolution and the convolution sum;
3. Answer the following questions in the lab report:
 - ① Write down the formula and computational steps for convolution integral;
 - ② Express the convolution of a signal $f(t)$ with an impulse function.

III. Experimental Principles

1. Convolution

The convolution integral method allows the zero-state response of a linear time-invariant (LTI) system to be computed using its impulse response. The zero-state response $r(t)$ is given by the convolution of the input signal $e(t)$ and the system's impulse response $h(t)$, expressed as:

$$r(t) = e(t) * h(t) \quad (3.1)$$

The computation of the convolution integral involves four steps: Folding, Shifting, Multiplication, Integration. The key to evaluating the integral is determining the correct integration limits.

Given two functions $f_1(t)$ and $f_2(t)$, the integral $f(t) = \int_{-\infty}^{\infty} f_1(\tau) f_2(t - \tau) d\tau$ is called the convolution integral of $f_1(t)$ and $f_2(t)$, denoted as

$$f(x) = f_1(t) * f_2(t)$$

2. Algebraic Properties of Convolution

- (1) Commutative Property: $f_2(t) * f_1(t) = f_1(t) * f_2(t)$
- (2) Distributive Property: $f_1(t) * [f_2(t) + f_3(t)] = f_1(t) * f_2(t) + f_1(t) * f_3(t)$

The distributive property applies to parallel systems: the overall impulse response of a parallel connection equals the sum of the individual subsystem impulse responses.

- (3) Associative Property: $[f_1(t) * f_2(t)] * f_3(t) = f_1(t) * [f_2(t) * f_3(t)]$.

The associative property applies to cascaded systems: the overall impulse response of a series connection equals the convolution of the individual subsystem impulse responses.

IV. Experimental Procedure

Rectified Sine Wave Input

Generate a sine wave from analog output DAC-0. Let the analog signal pass through the SAMPLE/HOLD (S/H) module to create a discrete pulse sequence. Note that the pulse generator and DAC share the same internal clock. Therefore, the oscilloscope display shows no timing delay. Refer to Figures 3.3 and 3.4 for the schematic and connection diagram.

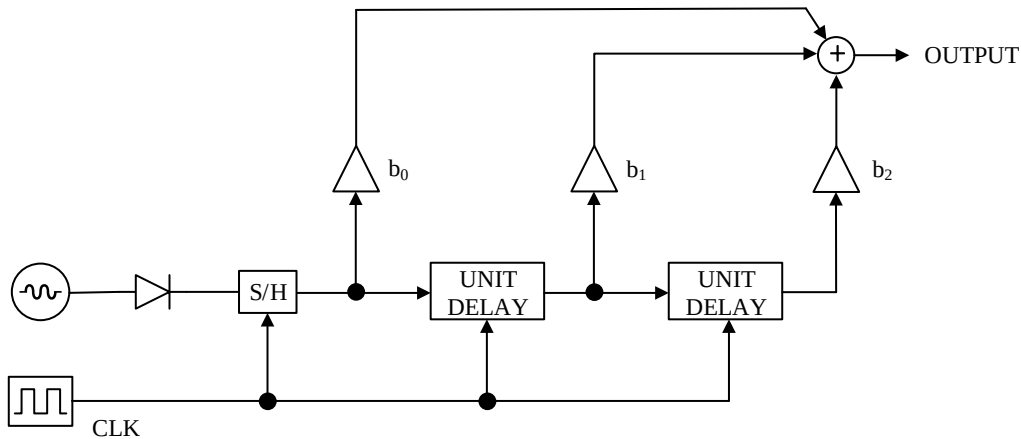


Figure 3.3. System block diagram with pulse sequence as input.

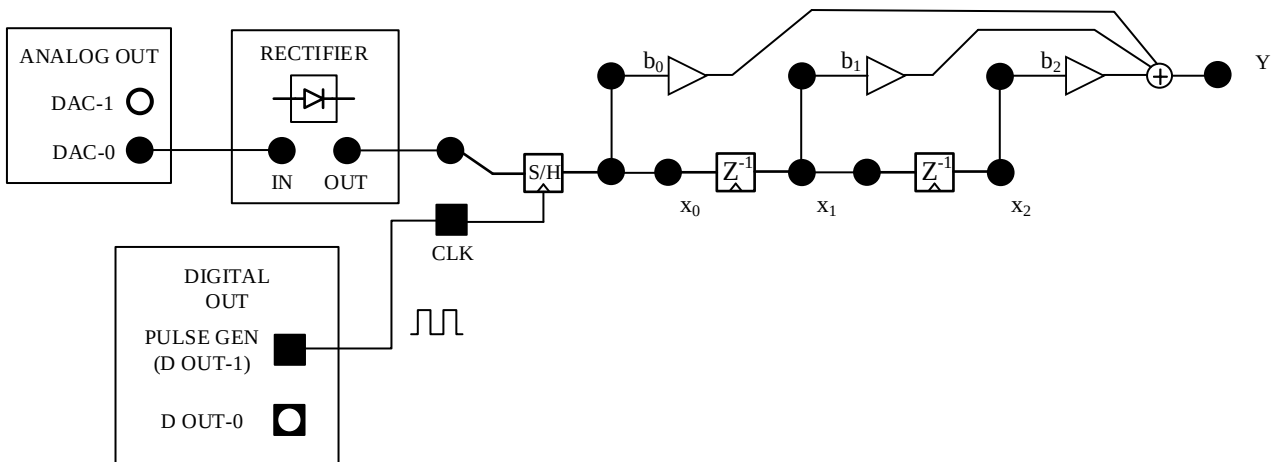


Figure 3.4. Connection diagram with rectified sine wave as input

Set the following parameters (Using Lab5):

- ① Pulse generator: Frequency: 800 Hz, Duty cycle: 0.5 (50%)
- ② Oscilloscope: Time base: 10 ms/div, Trigger: Rising edge on CH0, Trigger level: 1 V

Use **CH0** to observe the **DAC-0** output. Confirm that the sine wave input to the rectifier has a frequency of 100 Hz and a peak amplitude of 2 V. Observe the input and output of the S/H module using the oscilloscope. Record the waveforms.

Question 1

Measure and record the amplitude of the half-wave rectified signal. Explain why the amplitude is

reduced.

Question 2

What happens to the sampling process if the pulse generator frequency is changed? Is a higher frequency always better? Why or why not?

Keep the gains fixed at $b_0 = 0.3$, $b_1 = 0.5$, $b_2 = -0.2$. Use the output of the S/H module as the system input and the adder output Y as the system output. Observe both signals on the oscilloscope. Measure and record the amplitude of each pulse in Table 3.1. The index represents the pulse number, with the first pulse in each period being labeled as index 0. Count the number of non-zero outputs and mark them on the recorded waveform.

Table 3.1. Verification of system additivity

Amplitude Index	Input(V)	Output(V)			
		$b_0 = 0.3,$ $b_1 = 0.5,$ $b_2 = -0.2$	$b_0 = 0.3,$ $b_1 = 0,$ $b_2 = 0$	$b_0 = 0,$ $b_1 = 0.5,$ $b_2 = 0$	$b_0 = 0,$ $b_1 = 0,$ $b_2 = -0.2$
0					
1					
2					
3					
4					
5					
6					
7					
8					

Change the gain values of b and record the corresponding pulse amplitudes in Table 3.1.

V. Important Notes

1. General safety and setup procedures are the same as in Experiment 2.
2. When verifying additivity, carefully identify the index of each measured pulse. Plan the measurement before recording data.